

Andi Zhang

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Education

University of Cambridge

PhD in Computer Science

Oct. 2019 - Mar. 2024

- Supervised by Dr. Damon Wischik. Study Bayesian neural network, Probabilistic generative models and applications about robustness of deep learning models (adversarial examples, OOD detection).

MAst in Mathematical Statistics (Part III)

Oct. 2018 - Jun. 2019

- Mathematical Tripos Part iii. Relevant modules:
Statistical Learning in Practice, Convex Optimization, Astrostatistics
Bayesian Modelling and Computation, Modern Statistical Methods

MPhil in Advanced Computer Science

Oct. 2017 - Jun. 2018

- Relevant modules:
Machine Learning for Natural Language Processing, Algebraic Path
Deep Learning for Natural Language Processing, Advanced Functional Programming
Probabilistic Machine Learning

University of Manchester

BSc (Hons) in Computer Science and Mathematics

Sep. 2014 - Jun. 2017

- Relevant modules:
Machine Learning, Quantum Computing, Compilers
Convex Optimization, Probability, Statistical Methods
Functional Analysis, Topology, Number Theory

Experience

Kirin AI Group, Huawei

Cambridge, United Kingdom

Research Intern (Part-time) | Mentor: Mr. Arnau Raventos

Nov. 2021 - Oct. 2022

- Worked on video frame interpolation and AI global illumination.

Noah's Ark Lab, Huawei

Shenzhen / Beijing, China

Research Intern | Mentor: Dr. Yitong Sun

Sep. 2020 - Sep. 2021

- Worked on deep learning uncertainty, out-of-distribution detection and lossless compression.

University of Cambridge

Cambridge, United Kingdom

Course Supervisor (Teaching Assistant)

Oct. 2019 - Jun. 2023

- Supervised IA Algorithms (2020 Lent, 2023 Lent), IA Machine Learning in Real World (2020 Lent), IA Introduction to Probability (2020 Easter, 2021 Easter), IB Further Java (2020 Michaelmas), IB Data Science (2022 Michaelmas), IB Further Graphics (2022 Michaelmas) and II Machine Learning and Bayesian Inference (2023 Lent)

MRC Biostatistics Unit, University of Cambridge

Cambridge, United Kingdom

Research Assistant | Supervisor: Dr. Sofia Villar and Dr. David Robertson

Jul. 2019 - Sep. 2019

- Built a simulation system for multi-armed bandit problem (MABP) in clinical trial.
- Presented a new test statistic for binary response MABP.

Tencent (WeChat AI)

Beijing, China

R & D Intern | Mentor: Dr. Jie Zhou

Jul. 2018 - Sep. 2018

- Concentrated on reinforcement learning and evolution strategy.

Amazon (Alexa)

Cambridge, United Kingdom

Applied Scientist Intern | Mentor: Dr. Emilio Monti

Jun. 2017 - Sep. 2017

- Concentrated on Neural Machine Translation (Sequence to sequence model).

Robotics Institute, Carnegie Mellon University

Pittsburgh, United States

Student Intern | Supervisor: Dr. Stelian Coros

Jun. 2016 - Sep. 2016

- Assisted in implementing general stance controller for the 3D printable robotic creatures.
- Helped reproduce the prototype animation system.

Publications

Constructing Semantics-Aware Adversarial Examples with Probabilistic Perspective <i>Andi Zhang, Damon Wischik</i> https://arxiv.org/abs/2306.00353	Arxiv Preprint
SR-OOD: Out-of-Distribution Detection via Sample Repairing <i>Rui Sun*, Andi Zhang*, Haiming Zhang*, Yao Zhu, Ruimao Zhang, Zhen Li</i> https://arxiv.org/abs/2305.18228	Arxiv Preprint
Falsehoods that ML researchers believe about OOD detection <i>Andi Zhang, Damon Wischik</i> In ML Safety Workshop, 36th Conference on Neural Information Processing Systems	NeurIPS 2022 Workshop
Out-of-Distribution Detection with Class Ratio Estimation <i>Mingtian Zhang*, Andi Zhang*, Tim Z. Xiao, Yitong Sun, Steven McDonagh</i> In ML Safety Workshop, 36th Conference on Neural Information Processing Systems	NeurIPS 2022 Workshop
On the Out-of-distribution Generalization of Probabilistic Image Modelling <i>Mingtian Zhang*, Andi Zhang*, Steven McDonagh</i> In Neural Information Processing Systems annual meeting, 2021	NeurIPS 2021
Measuring Uncertainty in Signal Fingerprinting with Gaussian Processes Going Deep <i>Ran Guan, Andi Zhang, Mengchao Li, Yongliang Wang</i> In International Conference on Indoor Positioning and Indoor Navigation, 2021	IPIN 2021
Towards dynamic and scalable active learning with neural architecture adaption for object detection <i>Fuhui Tang, Dafeng Wei, Chenhan Jiang, Hang Xu, Andi Zhang, Wei Zhang, Hongtao Lu, Chunjing Xu</i> In The British Machine Vision Conference, 2021	BMVC 2021
Solipsistic Reinforcement Learning <i>Mingtian Zhang*, Peter Noel Hayes*, Tim Z. Xiao, Andi Zhang, David Barber</i> In Self-Supervision for Reinforcement Learning Workshop - International Conference on Learning Representations, 2021	ICLR 2021 Workshop

Honers & Awards

Top Reviewer (10%) of AISTATS 2023	Feb. 2023
The Williams/Kilburn Medal for outstanding final year student School of Computer Science, University of Manchester The medal is awarded on an occasional basis to students of exceptional distinction whose performance throughout their undergraduate course has been of outstanding merit.	Jul. 2017
University awards for outstanding academic achievement University of Manchester The University will consider the top 0.5% of undergraduates (nominated by each School) for this award.	Jul. 2017